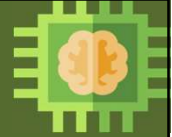


Elective Course

Course Code: CS4103

Autumn 2026-27

**Lecture #04**

Artificial Intelligence for Data Science

Module-2: PROBLEM SOLVING BY SEARCH

Problem formulation, Concept of state space search

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Problem Solving Agent



- **Goal-based agent**
 - Agent needs to achieve certain goals
 - Decides what to do by finding sequences of actions that lead to desirable states/goals
 - Many problems can be represented as a set of rules of how one state is transformed to another. The goal-based agent must choose a sequence of action to achieve the desired goal
 - Must have a model of how the world evolves in response to actions
 - Optimal: Achieve goal at least cost

Problem Solving by Search



Formulate Problem

- **Initial State:** The description of the starting configuration
 - Each state is an **abstract representation** of the agent's environment and its configuration
- **Actions/Operators:** Takes the agent from one state to another state. A state can have a number of successor states.
- **Successor Function** (represented as **Transition Model** or Conditions of applicability)
 - Applied on a given state-action pair; returns a **successor state** which is reachable from the given state by applying the action.
 - **State Space:** The set of all the states reachable from an initial state
- **Goal test:** determines whether a given state is a **goal state**
- **Path Cost:** Usually, **path cost** is the sum of step cost. Step cost is the cost of transition from a state to its successor

Search:

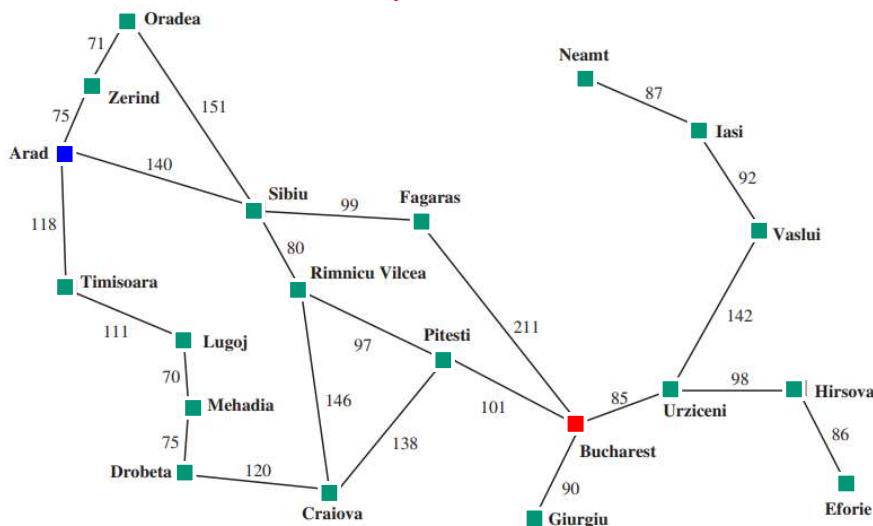
- Is the process of imagining sequences of actions/operators applied to the initial state, and checking which sequence reaches a goal state
- A **solution** is a **sequence of actions** leading from the initial state to a goal state
- Optimal solution: Least-cost solution

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Example Problem: Traveling in Romania



Currently in Arad. Find a route to drive to Bucharest.



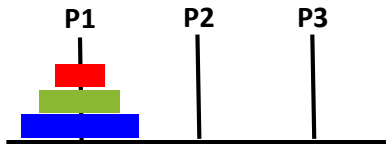
- **State space:**
 - Cities
- **Initial state:**
 - IN(Arad)
- **Actions:**
 - Drive between cities
- **Transition model:**
 - Reach adjacent city
- **Action cost/Step cost:**
 - Road distance from s to its successor s'
- **Goal test:**
 - $s = \text{IN}(\text{Bucharest})?$
- **Solution?**

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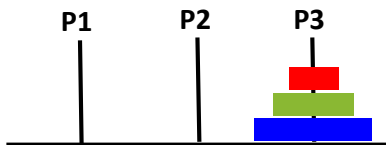
Problem: Three-Disk Tower of Hanoi Puzzle



• Start State



• Goal State

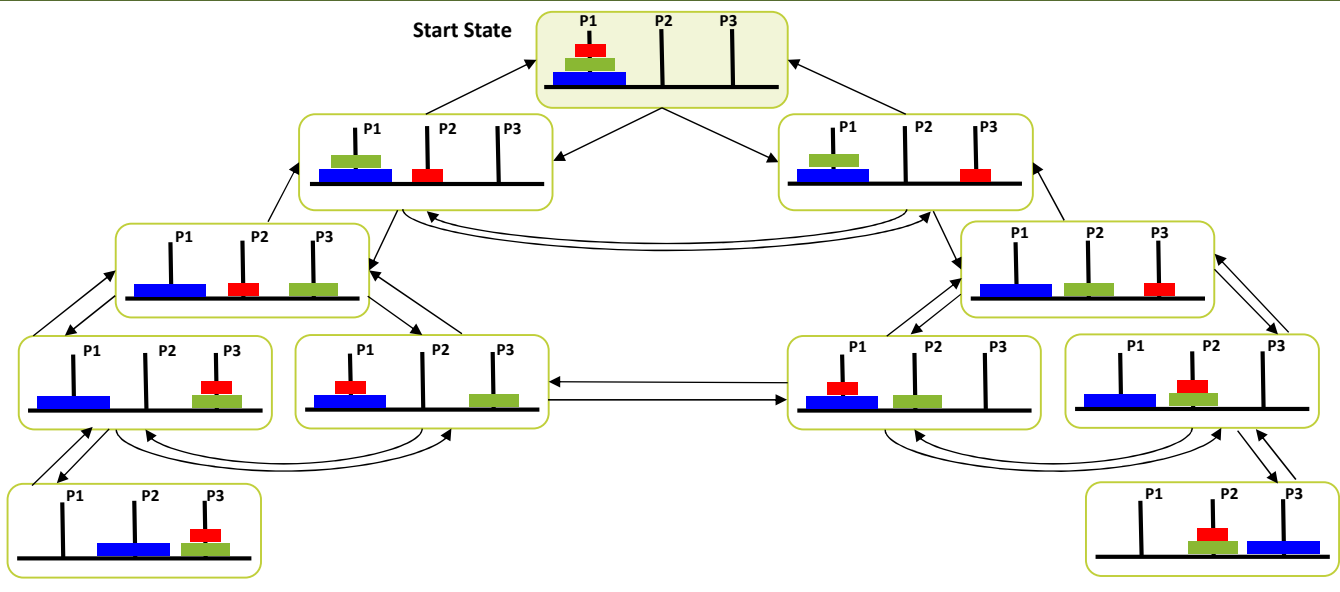


• Rules:

- Only one disk may be moved at a time.
- Each move involves taking disk on the top of a peg and place it on the top of another peg.
- A disk of a larger diameter should never be placed on the top of a disk of smaller diameter.

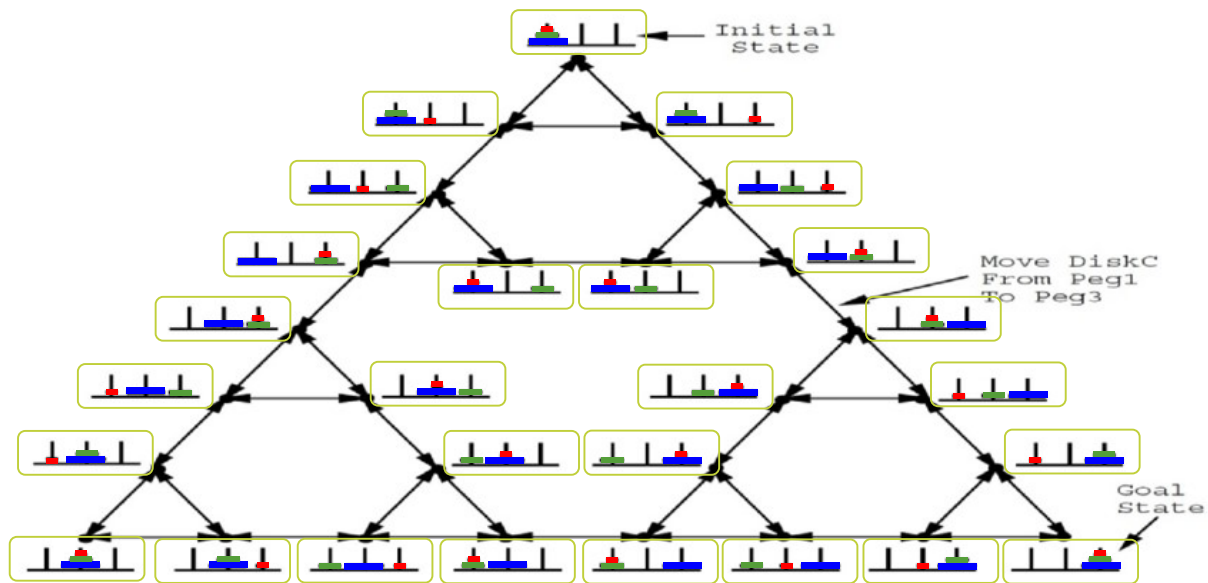
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State Space for the Three-Disk Tower of Hanoi Puzzle



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State Space for the Three-Disk Tower of Hanoi Puzzle

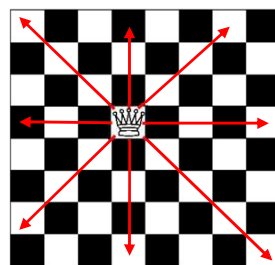
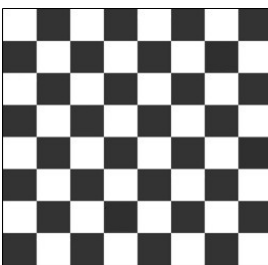


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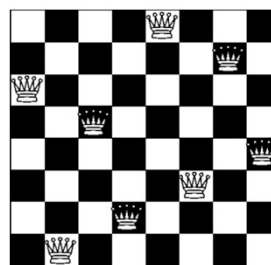
8 Queens Problem



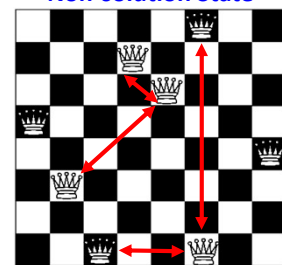
- Place 8 queens on chessboard so that no two queens are in the same row, column, or diagonal



Goal State



Non-solution State

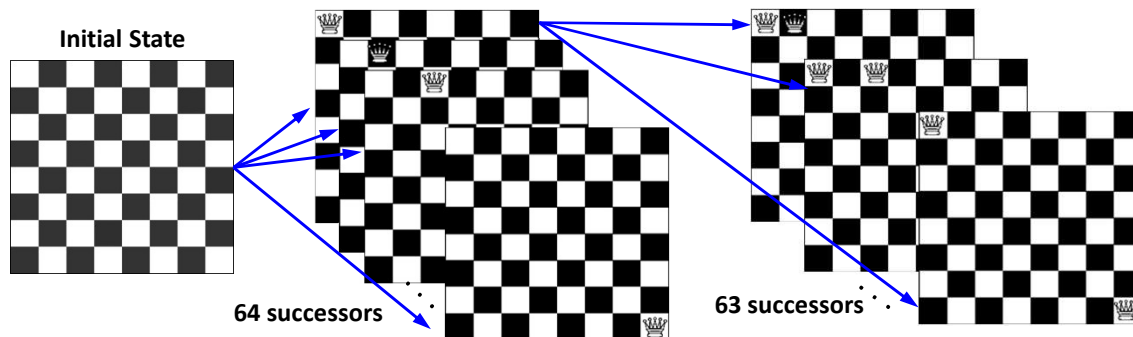


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Formulation-1 for 8-Queen Problem



- **States:** Any arrangements of 0 to 8 queens on the board
- **Initial State:** 0 queens on the board
- **Actions:** Add a queen to any empty square
- **Successor Function:** Returns the board with a queen added to the specified square.
- **Goal Test:** 8 queens on the board and none are attacked

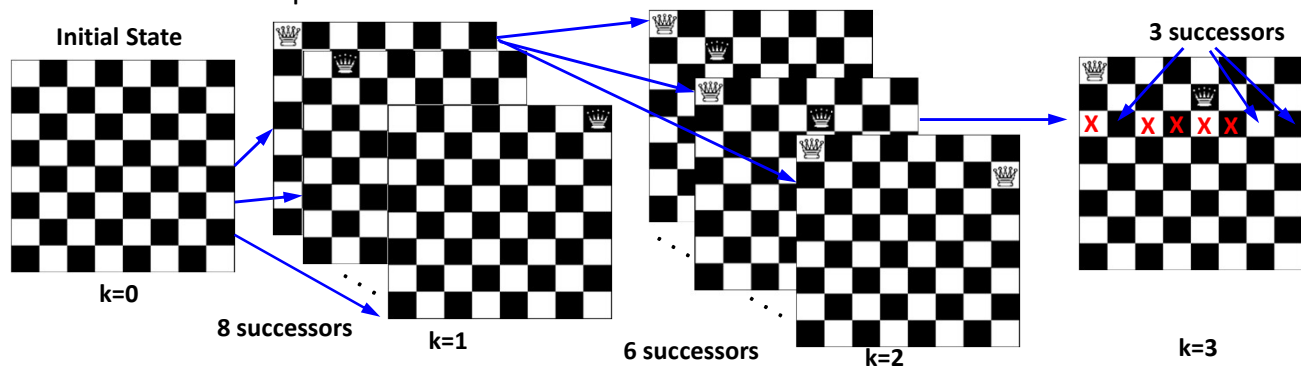


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Formulation-2 for 8-Queen Problem



- **States:** Any arrangement of k queens in the first k rows such that none are attacked
- **Initial State:** 0 queens on the board
- **Actions:** Add a queen to the $(k+1)$ -th row so that none are attacked
- **Successor Function:** Returns the board with a queen added to the specified square.
- **Goal Test:** 8 queens on the board and none are attacked



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Formulation: 8 Puzzle Problem



- **States**
 - A description of each of the eight tiles in each location that it can occupy
- **Operators/Actions**
 - The blank moves left, right, up or down (ML, MR, MU, MD)
- **Transition model**
 - Given a state and action, this returns the resulting state;
- **Goal Test**
 - The current state matches a certain state (e.g. see the fig.)
- **Path Cost**
 - Each move of the blank costs 1

5	4	
6	1	8
7	3	2

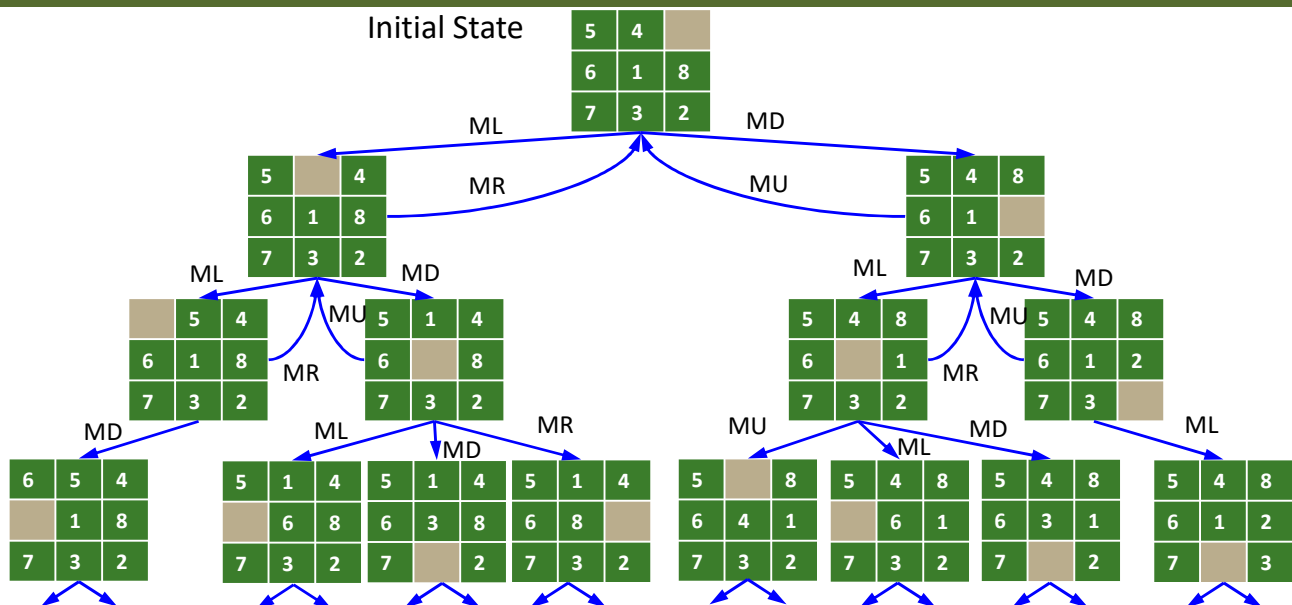
Initial State

1	4	8
5	6	2
7	3	

Goal State

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8 Puzzle Partial State Space



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State Space Search Problem



- **General problem:**

Find a path from a **start state** to a **goal state** given:

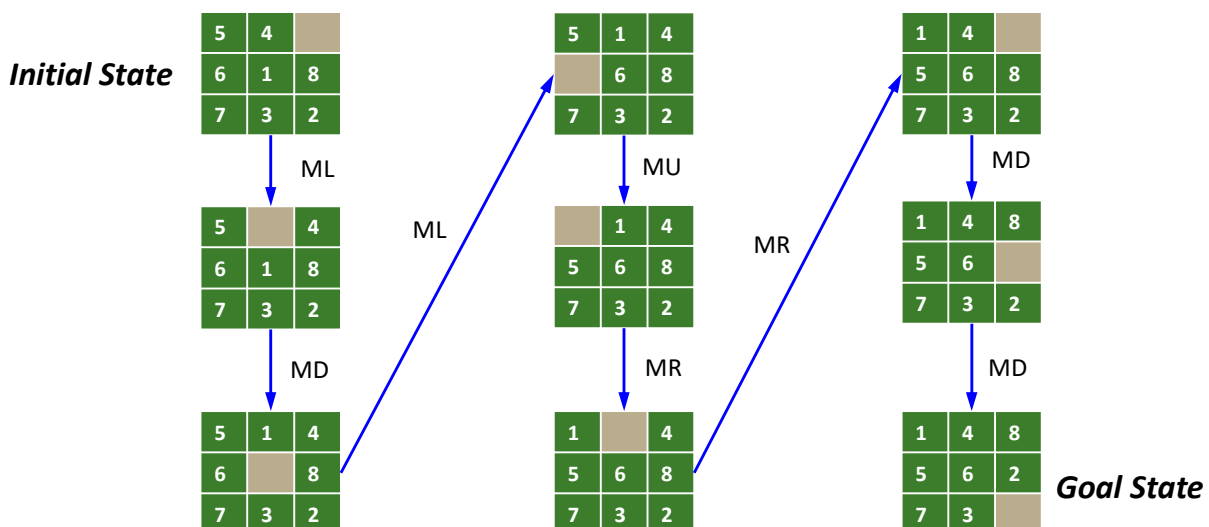
- Set of states
- Start/Initial State
- Goal state [test]
- Operators/Actions/
- Successor Function [and costs]

- **Output variants:**

- Find a **path** vs. a **least-cost path**
- Goal is completely specified, task is just to find the path
 - **Route Planning**
- Path doesn't matter, only finding the goal state
 - **N Queen**

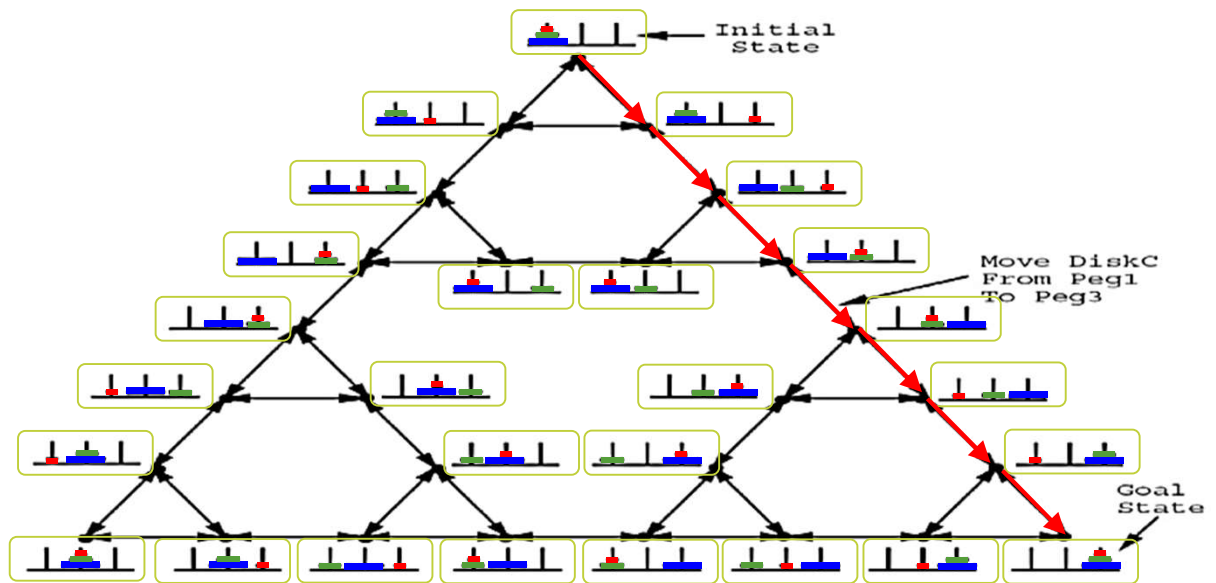
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8 Puzzle: A Solution Path



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Three-Disk Tower of Hanoi Puzzle: Solution Path

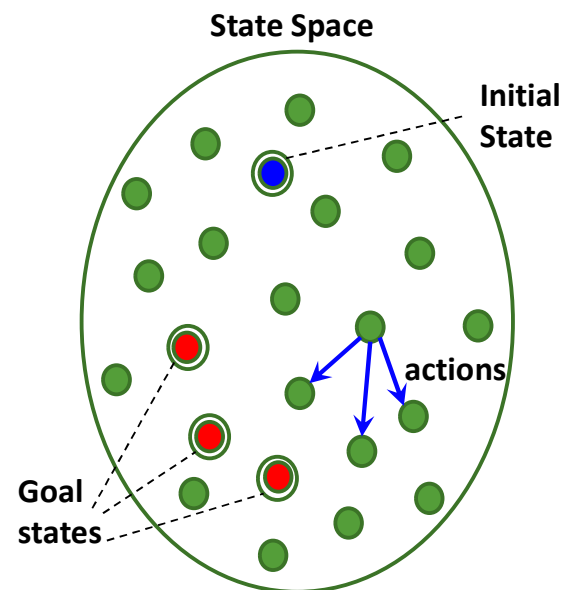


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Search Problem



- S : the full set of states
- s_0 : the initial state
- $A: S \rightarrow S$ set of operators/actions
- G : the set of final states. $G \subseteq S$
- **Search problem:** Find a sequence of actions which transforms the agent from the initial state to a goal state $g \in G$

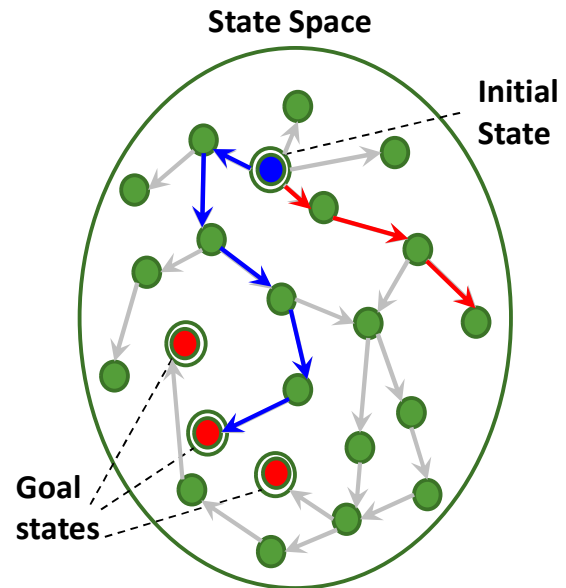


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Searching Process



- Check the current state
- Execute allowable actions to move to the next state
- Check if the new state is a solution state
 - If it is not, the new state becomes the current state and the process is repeated until a solution is found or the state space is exhausted



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Search Trees



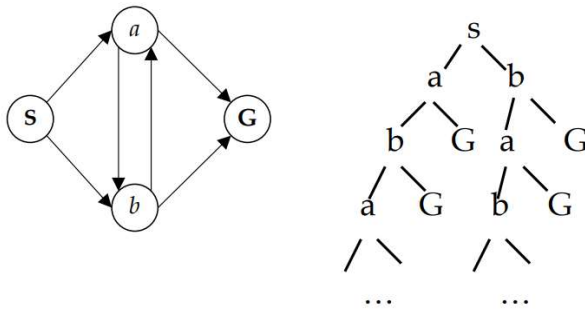
- Every search process can be viewed as traversal of tree structure in which each node represents a problem state and each arc represents a relationship between states represented by states it connects.
- **A search tree:**
 - **Root** = Start state
 - **Children** = successor states
 - **Edges** = actions and costs
 - **Path from Start to a node** is a “plan” to get to that state
 - **For most problems, we can never actually build the whole tree (why?)**

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State Space Graph versus Search Trees



Consider this 4-state graph:

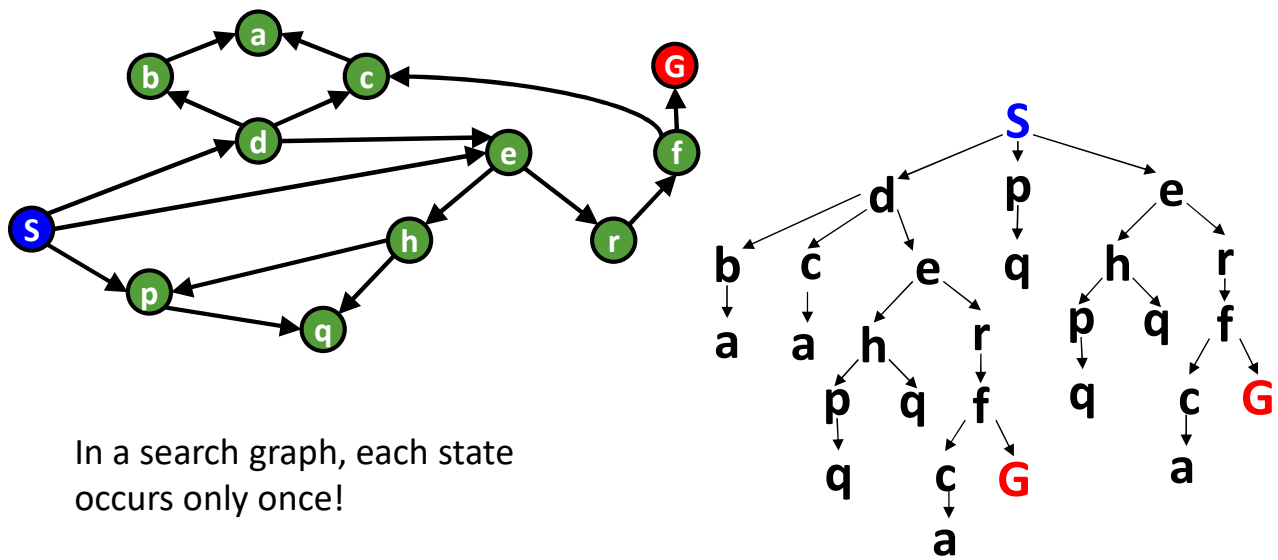


Important: Lots of repeated structure in the search tree!

slides adapted from Dan Klein, Pieter Abbeel at Berkeley and Dan Weld, Luke Zettlemoyer

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State Space Graph versus Search Trees



In a search graph, each state occurs only once!

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Basic Search Algorithm



Let L be a list containing the initial state (L=the fringe)

Loop

if L is empty return *failure*

Node $\leftarrow$ **select**(L)

if Node is a goal

then return Node

(*the path from initial state to Node*)

else apply all applicable operators to Node

and **merge** the newly generated states into L

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Search Strategy



- **Blind/Uninformed Search**
 - Depth first search
 - Breadth first search
 - Iterative deepening search
 - Uniform-Cost search
- **Informed/Heuristic Search**
 - Best First Search
 - A*,
 - Hill climbing
 - Simulated Annealing

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Home Assignments

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Problem-1: Missionaries and Cannibals Problem



- Missionaries and Cannibals is a problem in which 3 missionaries and 3 cannibals want to cross from the left bank of a river to the right bank of the river. There is a boat on the left bank, but it only carries at most two people at a time (and can never cross with zero people). If cannibals ever outnumber missionaries on either bank, the cannibals will eat the missionaries.
- A state can be represented by a triple, $(m \ c \ b)$, where m is the number of missionaries on the left, c is the number of cannibals on the left, and b indicates whether the boat is on the left bank or right bank.



Initial State

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Problem-1: Missionaries and Cannibals Problem



- For example, the **initial state** is (3 3 L) and the **goal state** is (0 0 R).
- **Operators are:**
 - MM: 2 missionaries cross the river
 - CC: 2 cannibals cross the river
 - MC: 1 missionary and 1 cannibal cross the river
 - M: 1 missionary crosses the river
 - C: 1 cannibal crosses the river

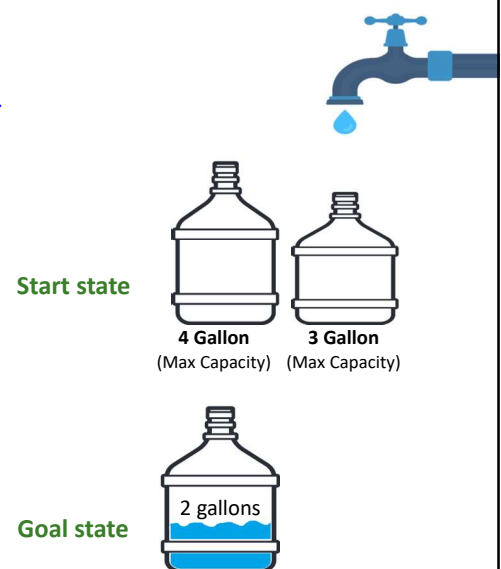
Draw a diagram showing all the legal states and transitions from states corresponding to all legal operations.

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Problem-2: Water Jug Problem



- You have a 4-gallon and a 3-gallon water jug
- You have a faucet with an unlimited amount of water
- You need to get exactly 2 gallons in 4-gallon jug
- **State representation:** (x, y)
 - x: Contents of 4-gallon jug
 - y: Contents of 3-gallon jug
- **Start state:** (0, 0)
- **Goal state:** (2, n)



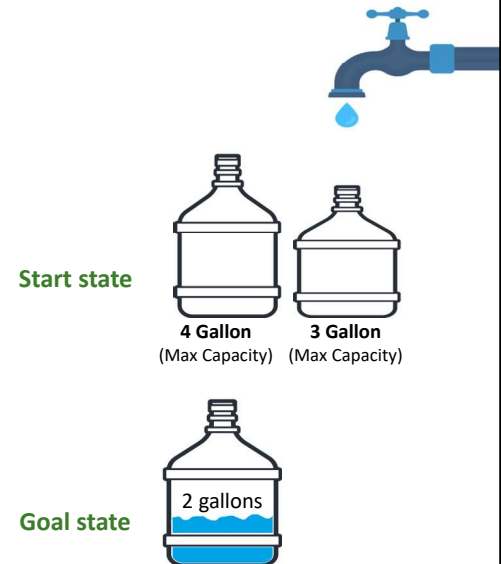
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Problem-2: Water Jug Problem



- **Operators**

1. Fill 3-gallon from faucet
2. Fill 4-gallon from faucet
3. Fill 3-gallon from 4-gallon
4. Fill 4-gallon from 3-gallon
5. Empty 3-gallon into 4-gallon
6. Empty 4-gallon into 3-gallon
7. Dump 3-gallon down drain
8. Dump 4-gallon down drain



Draw a diagram showing all the legal states and transitions from states corresponding to all legal operations.

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Questions?

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